Fundamentals of the Night Sky

Prepared By: Bijaya Luitel

Agenda



BASIC TERMS



SOME HISTORY



IMPORTANT CONSTELLATIONS



DETERMINING LOCATION



AN EXTREME SCENARIO!



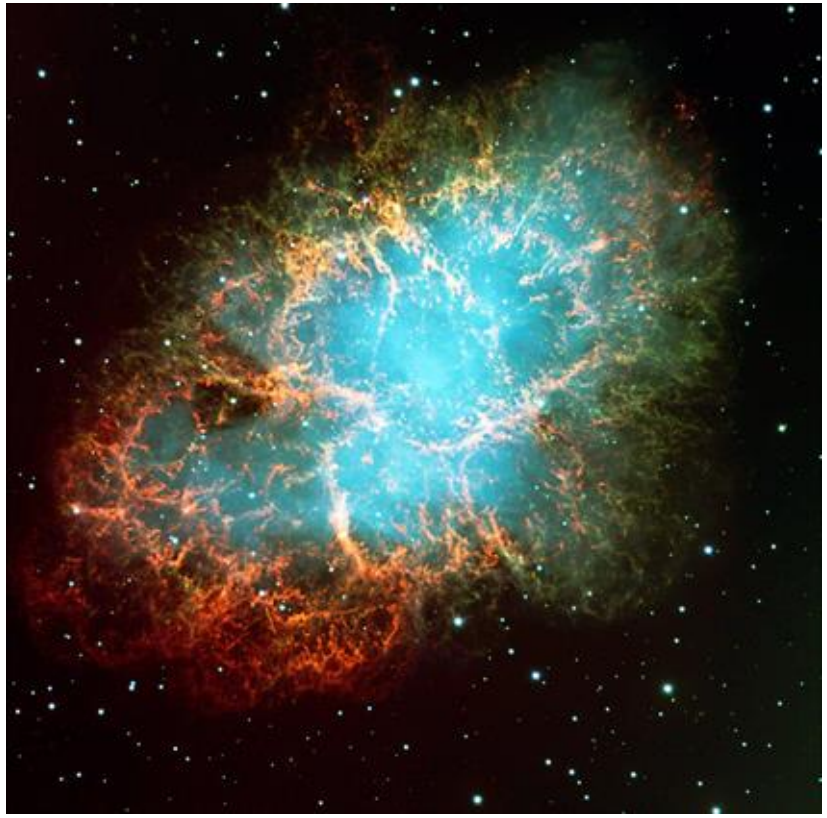
TAKING PICTURES



Picture credit:
https://www.environment-textures.com/photos/show/id/110777



Picture credit:
https://www.reddit.com/r/pics/comments/92f0b1/long_exposure_shot_that_shows_the_milky_way_edges/



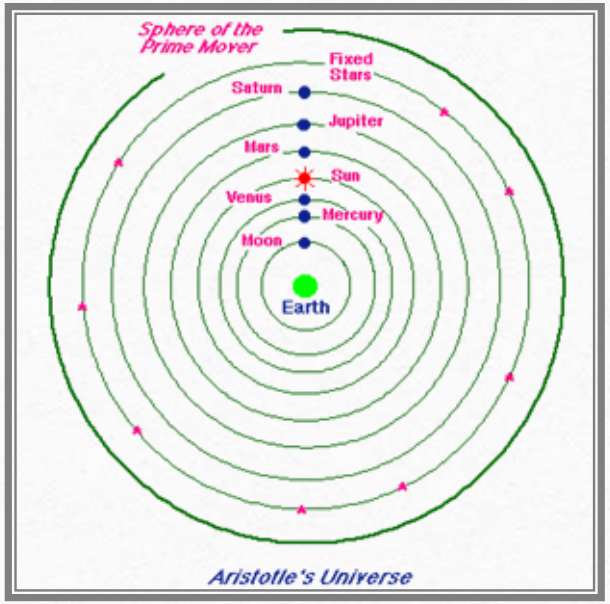
The Crab Nebula in the constellation of Taurus

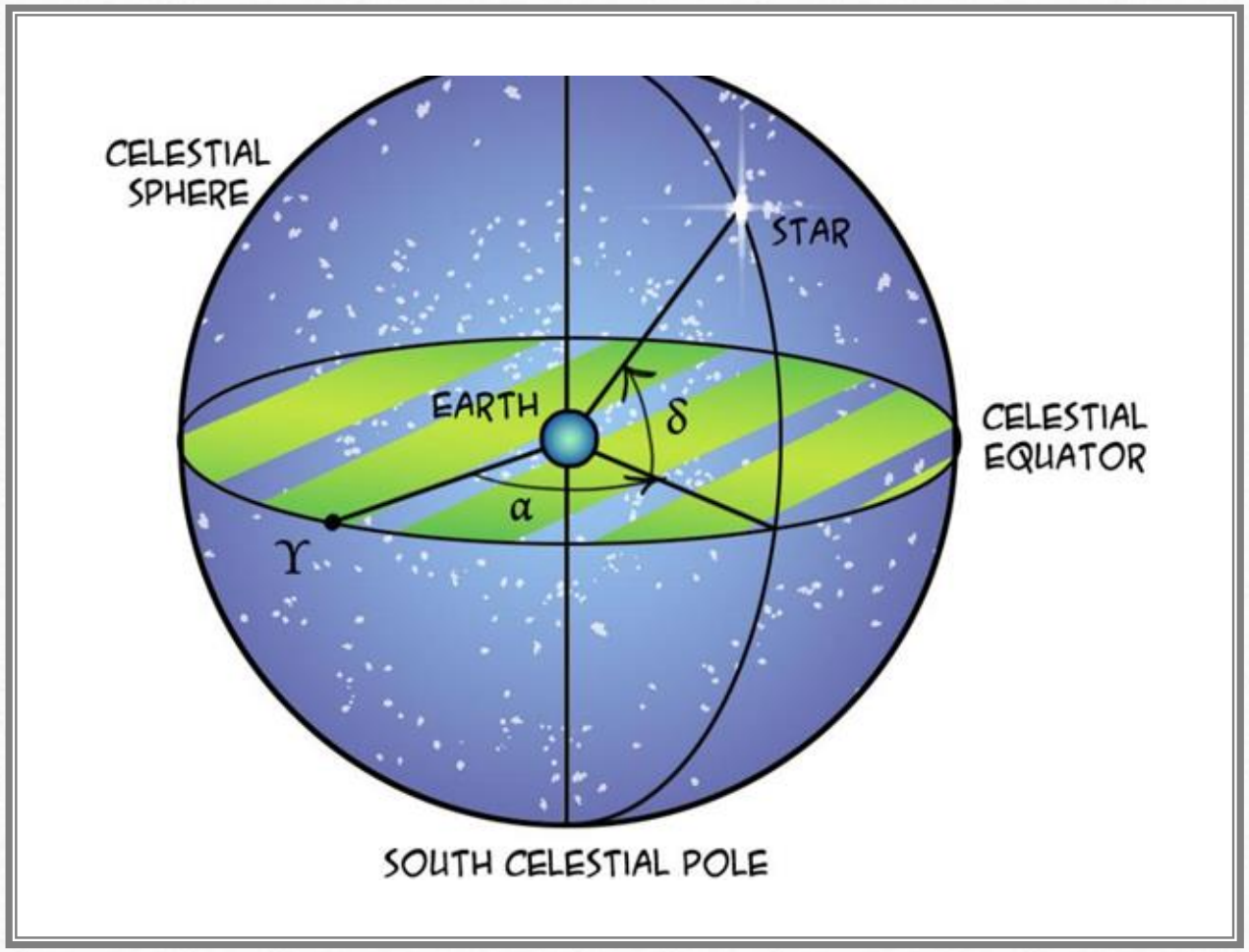
Why observe the sky?

- We can use it to navigate!
- We can tell what time it is!
- We can do interesting science by observing phenomena that take place immensely far away and back in time!
- AND… it is very RELAXING!!!

Earlier Ideas: Geocentrism

- The Earth was the center of the universe, with every observable heavenly body going around it.
- Of course, this was easy to justify given how the stars move across the sky in each day.
- However, with the invention of the telescope, and the mathematical simplicity of a heliocentric model of the Solar System, Geocentrism was abandoned.
- We now know that even the Sun is just a small part of the observable universe, which itself is predicted to have a finite size (~100 Billion Light Years in diameter).



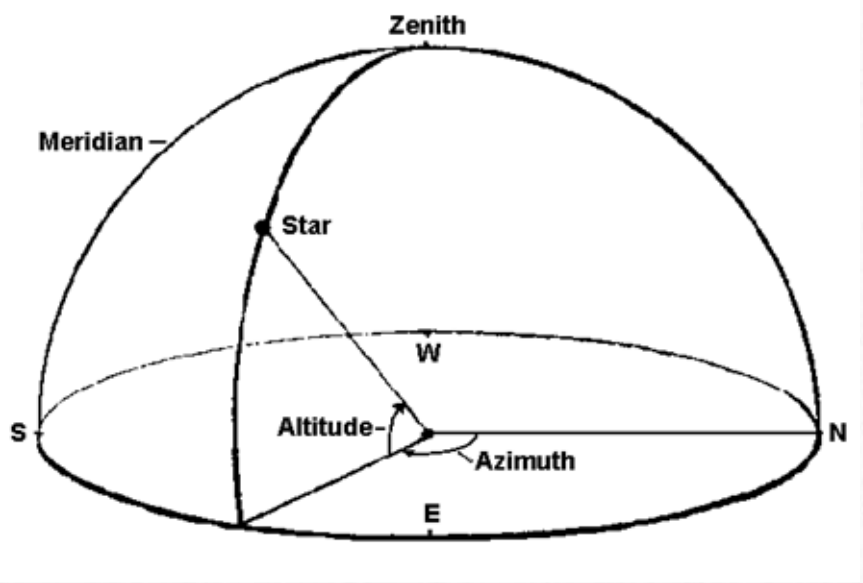


Relevance Today—The Equatorial Coordinate System

- Although the Earth is not at the center of the universe, for purposes of observation, it makes it easier to put the Earth at the origin of coordinate axes.
- Such a coordinate system, with the celestial equator at (0,0) is called the equatorial coordinate system.
- Like the latitude and longitude, we have two angles that measure the location of a star in this coordinate system, called the declination and the right ascension, respectively.
- The declination is analogous to the latitude and the Right Ascension is analogous to the longitude.

The Horizontal Coordinate System

- The Horizontal Coordinate system has two angles, the altitude and the Hour Angle.
- The altitude tells us how far up in the sky an object is and is measured in angles.
- The hour angle tells us how displaced an object is in comparison to the local meridian. The local meridian is the line that passes overhead for an observer.
- For observers far from the poles, the Azimuth is a good approximation—the Azimuth tells us how displaced an object is from North.



Some Important Constellations



The Orion Nebula can be seen by the naked eye!
Taken by: Shuin Jian, Processed by: Sean Neo

Orion

- It is the constellation of the hunter.
- It is the brightest constellation in the night sky, and can be seen from almost all parts of the world because of its relatively equatorial position.
- It is visible during the Northern Hemisphere Winter—in the month of December, it will have risen by the time the sun sets.
- Orion also contains a famous nebula that is visible with the naked-eye, shown here.



The "Winter Triangle", pictured is ***Orion*** and two bright stars, **Sirius** and **Procyon**.

Canis Major

- It is the constellation of the dog.
- It contains the brightest star in the sky, Sirius.
- It is located very close to Orion and so can be seen also in the winter months of the Northern Hemisphere.



The Zodiac

- It is the set of constellations through which the sun passes.
- There are *13* constellations of the zodiac.
- Because of an effect called precession—the movement of the Earth's axis—the sun's time within a constellation, as well as entry time changes over time.


what do you do?
i'm a astrophysicist.
i'm a Gemini.

Ursa Major

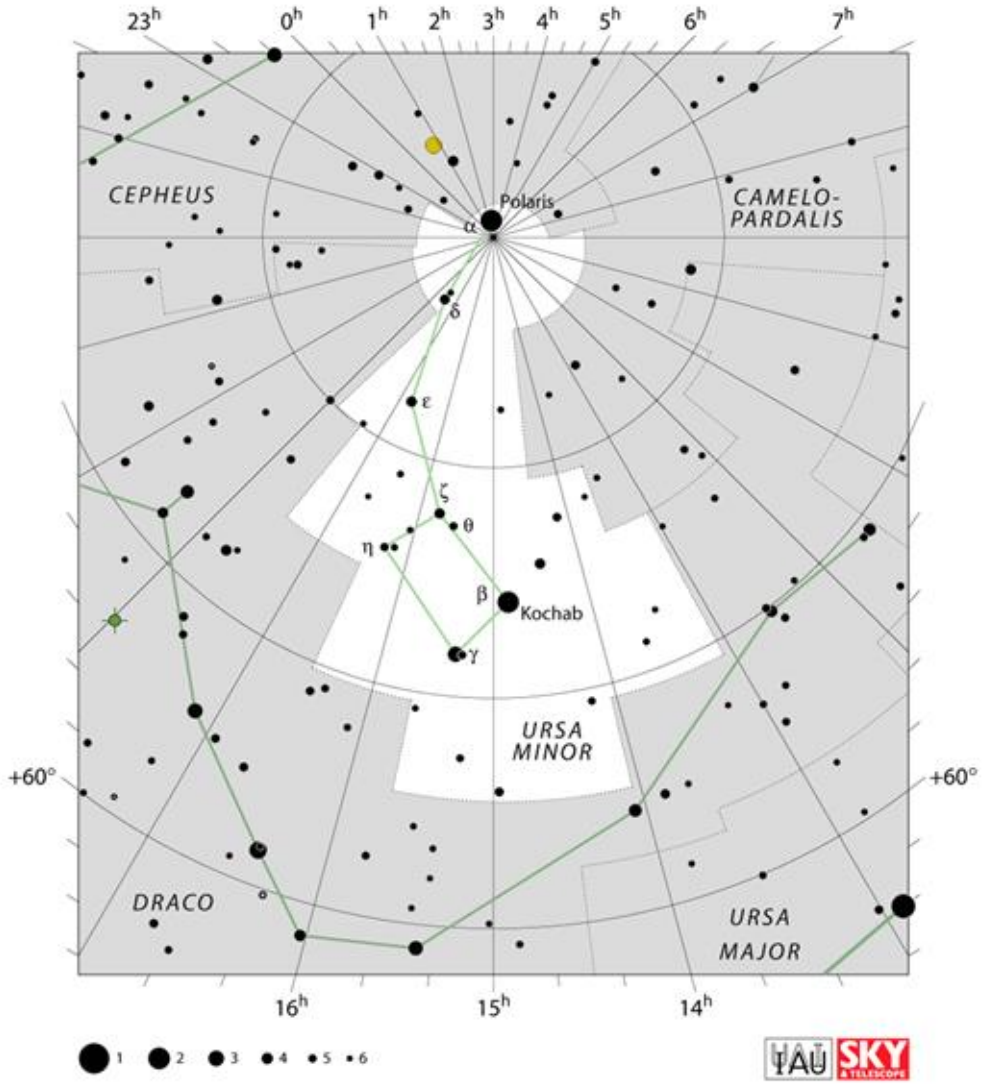
- It is the constellation of the great bear, and is famous for containing the Big Dipper.
- It is northerly constellation and so cannot be seen in its entirety from many parts in the Southern Hemisphere.
- In the Northern Hemisphere, parts of Ursa Major can be seen in almost all seasons.



The Big Dipper, part of ***Ursa Major***
Picture Credit: Ronald Zicone

Ursa Minor

- It is the constellation of the Little Bear.
- Like Ursa Major, Ursa Minor is a very northerly constellation.
- The star Polaris is located almost at the North Pole.
- In fact, this star can be used to determine a location's latitude—if the location is in the Northern Hemisphere.
- From the Northern Hemisphere, Polaris is always visible.



Centaurus

- Centaurus is the constellation of the Centaur.
- It is a relatively southerly constellation, so parts of it cannot be seen from many parts of the Northern Hemisphere.
- It contains Alpha Centauri, which is the closest star system to the solar system and is also the third brightest star in the sky.



The star cluster ***Omega Centauri***
Taken by: Shuin Jian
Processed by: Zhang Liu

Crux

- The Crux is the constellation of the Southern Cross.
- It is one of the most iconic constellations, with many countries in the Southern Hemisphere having some variant of the crux on their flag.
- It is also a very bright constellation, and can be seen almost year-round from locations in the Southern Hemisphere.



Crux in the night sky

Locating Oneself

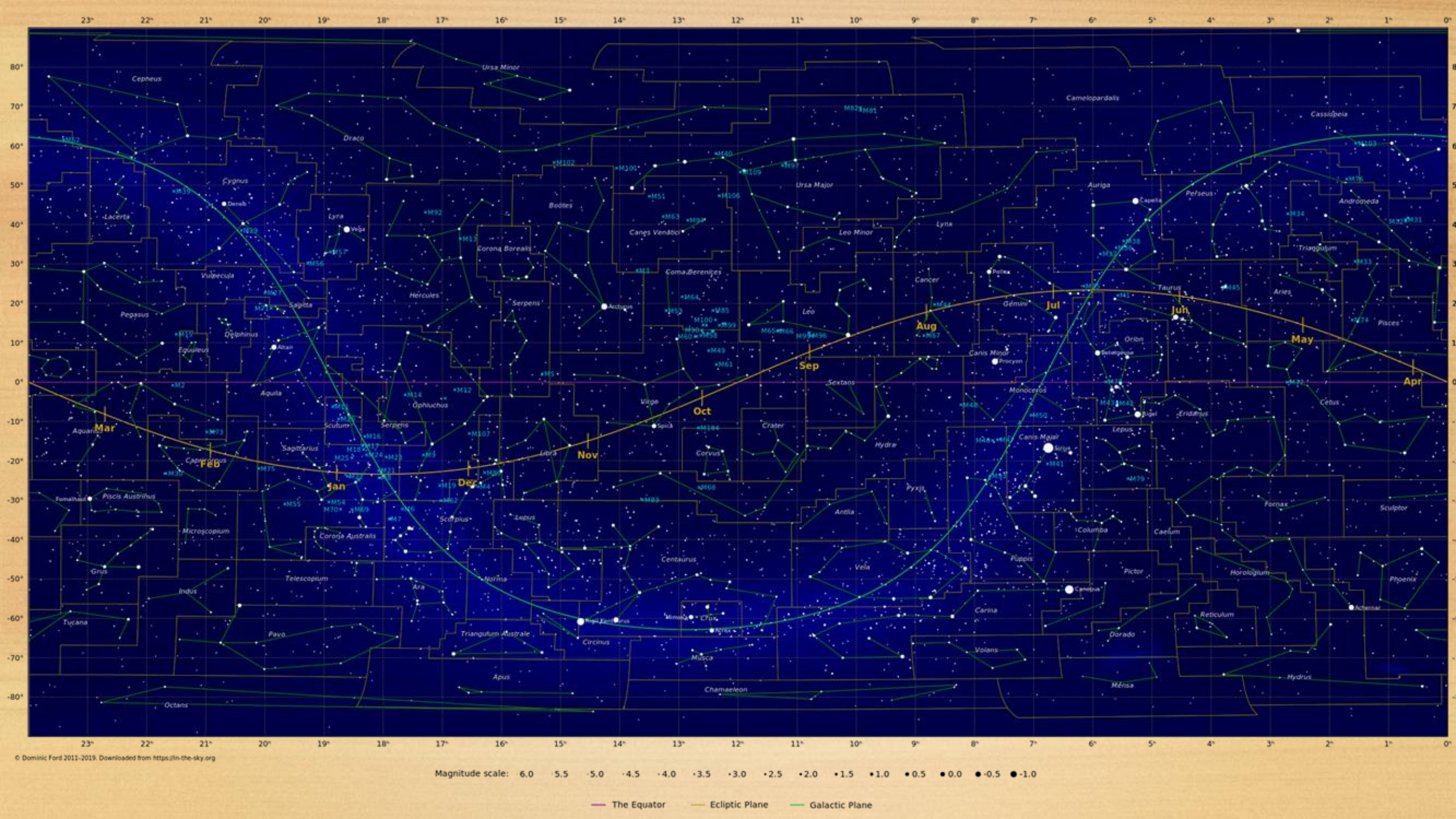
- Using the location of a few stars based on the coordinates mentioned earlier, we will be able to locate ourselves on the Earth.
- The easiest way to do so in the Northern Hemisphere is using the star Polaris.
- Polaris always appears due North, which you can use to determine the other cardinal points.
- You can also use Polaris to determine a location's latitude!



***The Pleiades*, in Taurus.**
Taken and processed by: Shuin Jian

Telling seasons

- One quick way to determine what season it is by looking at the night sky is observing the constellations that you can see.
- If you see Orion already high up in the sky towards the East by sunset time, its Northern Hemisphere winter.
- If you see Orion setting when the sun is setting, then its probably spring.
- Conversely, if you see the bright asterism the Summer Triangle high up in the sky at night, you can be pretty certain that it's summer.



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Magnitude scale: 6.0 5.5 5.0 4.5 4.0 3.5 3.0 2.5 2.0 1.5 1.0 0.5 0.0 -0.5 -1.0

The Equator — Ecliptic Plane — Galactic Plane

When knowing the stars is lifesaving—literally

"Escape"

Instruction

After being shipwrecked (by the theory paper), you are stranded on an unknown island at an unknown location. Luckily, behind the door is a satellite radio that you can use to escape the island. You must use any information available to you to find the correct combination to unlock the door.

Bolted on the door are three combination locks. Enter the correct 4-digit combination and the lock will open.

The information required on the three locks are:

A. The **Latitude** of this island (in "00LL" format, 0000 - 0090)
B. The **Longitude** of this island (in "0LLL" format, 0000 - 0180)
C. The **Month** of the observed sky above you (in "00MM" format, 0001 - 0012)

In order to solve this, the sky above you is simulated from the location of the island, but with time passing faster than real time. The *only* tools you are given are:

- An equatorial sky chart
- A clock showing the current time in GMT (remember, the time is sped up)
- Pencil and paper

The answers, in this case, are literally written in the sky.

You have 2 nights to find an escape. The game will end upon the second dawn.

Examples in Stellarium



Credit: Yuri Beletsky Nightscapes.

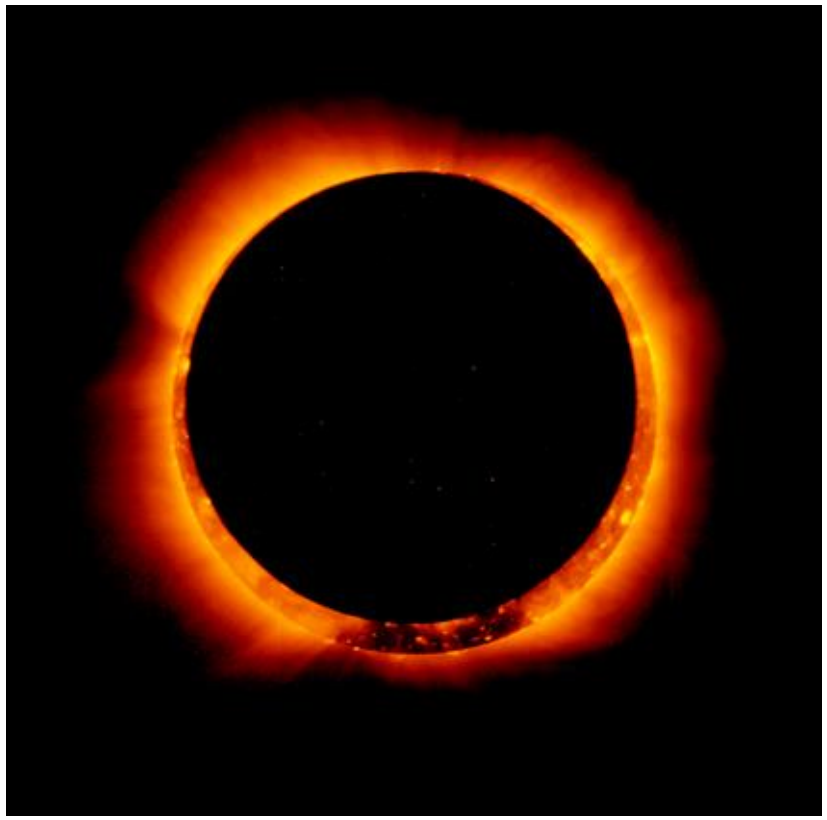


Image Credit: NASA/Hinode/XRT

Events to look forward to!

- Solar eclipse (annular) – December 26th
- Quadrantid Meteor shower ~ New Year's Day

Thank you!